Features

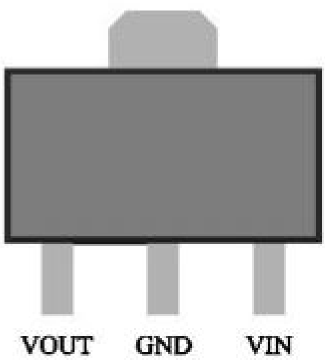
- Output Current of 100mA
- Thermal Overload Protection
- Short Circuit Protection
- Output transistor safe area protection
- No external components
- Package：SOT89-3 and TO92
- Output voltage accuracy: tolerance ±5%

General Description

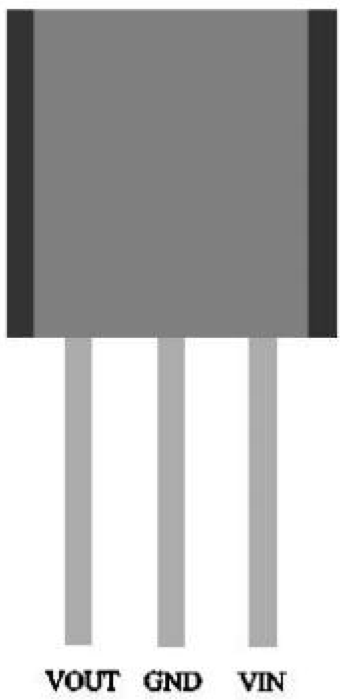
HX78LXXis three-terminal positive regulators. One of these regulators can deliver up to 100 mA of output current. The internal limiting and thermal -shutdown features of the regulator make them essentially immune to overload. When used as a replacement for a zener diode-resistor Combination, an effective improvement in output impedance can be obtained, together with lower quiescent current.

Pin Configuration

SOT89 (Top view)



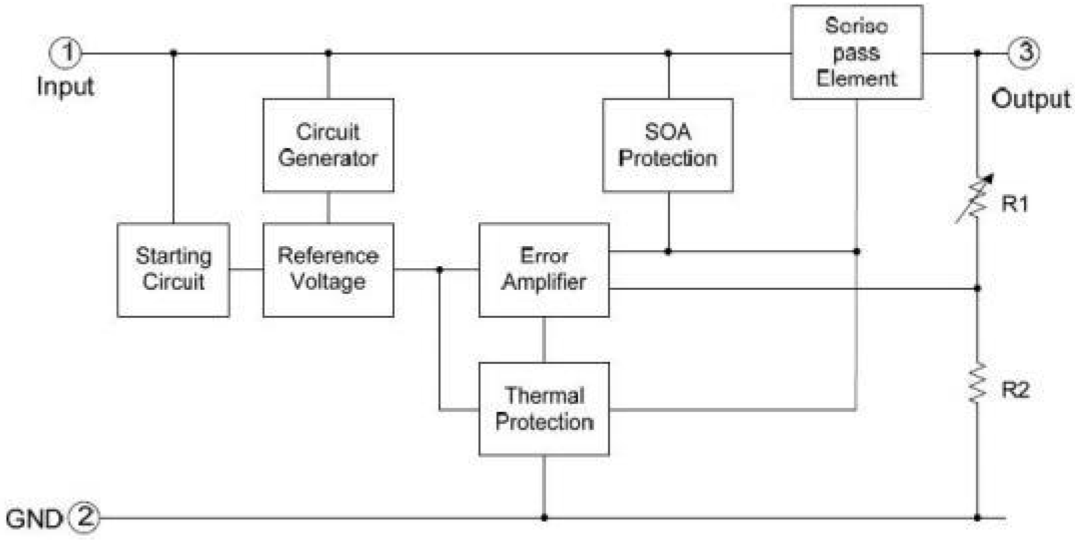
TO92 (Front view)



Selection Table

Part No.	Output Voltage	Package	Marking
HX78L05	5.0V	TO92 SOT89	
HX78L06	6.0V		
HX78L08	8.0V		
HX78L09	9.0V		
HX78L12	12V		

Block Diagram



Absolute Maximum Ratings（Ta=25℃）

Parameter	Rating	Unit
Input supply voltage: VIN	30	V
MAX. Output current:Iout	100	mA
MAX Power:Pmax	0.5	W
Maximum junction temperature:Tj	-25~125	℃
Storage temperature:Tstr	-55~125	℃
Soldering temperature and time	+260(Recommended 10S)	℃

Note: The absolute maximum ratings are rated values exceeding which the product could suffer physical damage. These values must therefore not be exceeded under any conditions.

Electrical Characteristics

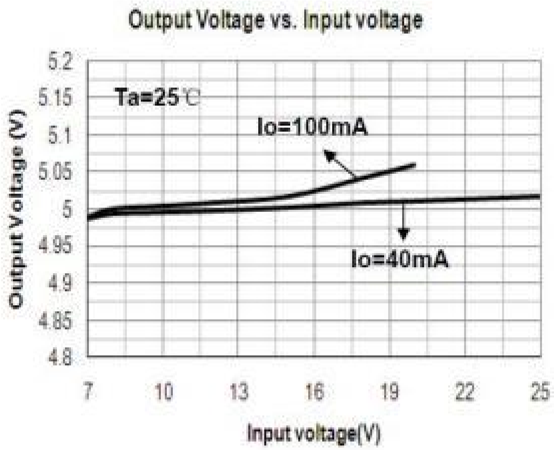
(Cin=0.33uF, Co=0.1uF, 0≤Tj≤125℃， unless otherwise noted)

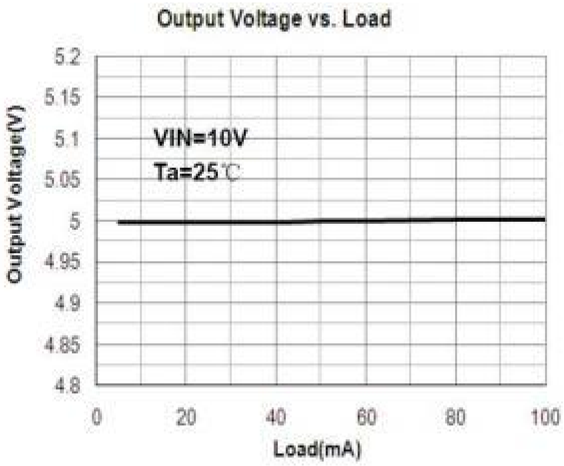
Parameter	Symbol	Conditions	Min.	Typ.	Max.	Unit
Output Voltage	Vout	Io=40mA, VIN=10V	0.964vout	vout	1.036vout	V
		Io=1mA~40mA VIN=7V~ 18V	0.96vout	vout	1.04vout	
		Io=1mA~ 10mA VIN=10V	0.95vout	vout	1.05vout	
Line Regulation	LNR	VIN=7V~ 18V, Io=40mA	-150	-	150	mV
		VIN=8V~ 18V, Io=40mA	-100	-	100	
Load Regulation	LDR	VIN=10V, Io=1mA~ 100mA	-100	-	100	mV
		VIN=10V, Io=1mA~40mA	-30	-	30	
Dropout Voltage	V_{DIF}	Tj=25℃， Io=100mA	-	2	-	V
Output noise Voltage	V_N	F=10Hz to 100KHz	-	40	-	uV/Vo
Ripple Rejection	PSRR	Tj=25℃， f=120Hz, Io=40mA, VIN=8V~20V	-	80	-	dB
Quiescent Current	I_Q	VIN=10V, IOUT=40mA	-	-	5.5	mA
Quiescent Current Change	$\triangle I_Q$	VIN=8V~ 18V, I_O=40mA	-1.5	-	1.5	mA
		VIN=10V, IOUT=1mA~40mA,	-0.1	-	0.1	

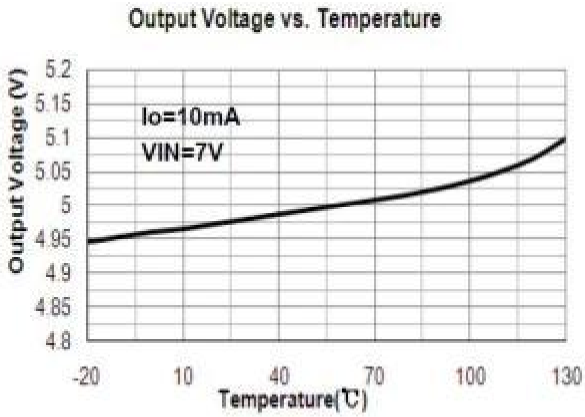
LNR: Line Regulation. The change in output voltage for a change in the input voltage. The measurement is made under conditions of low dissipation or by using pulse techniques such that the average chip temperature is not significantly affected.

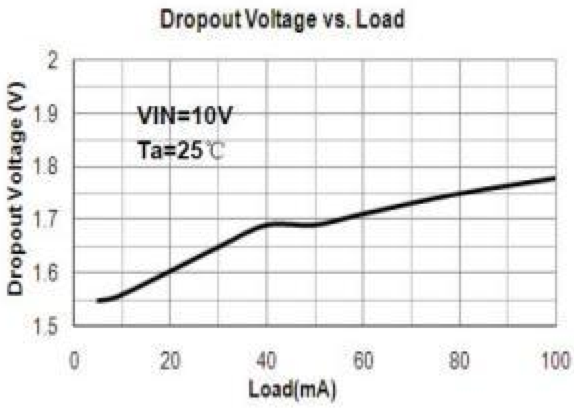
LDR: Load Regulation.The change in output voltage for a change in load current at constant chip temperature.

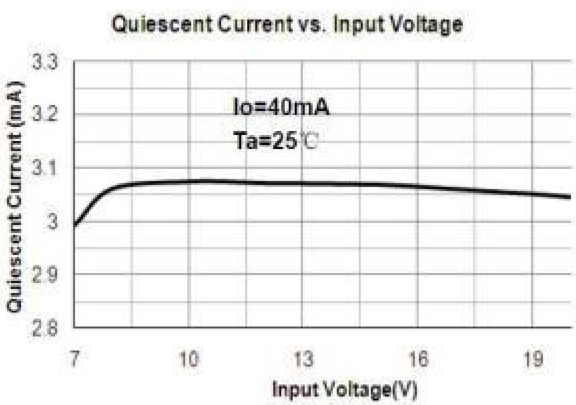
Typical Performance Characteristics

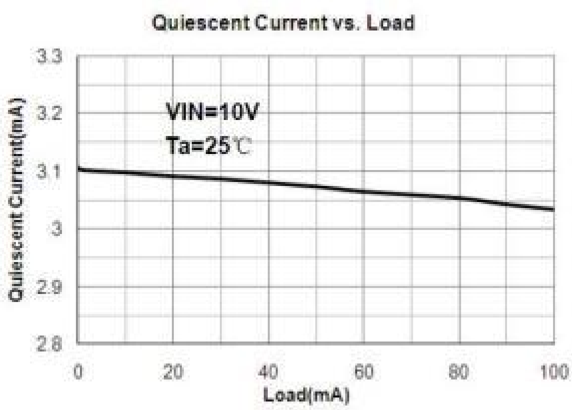












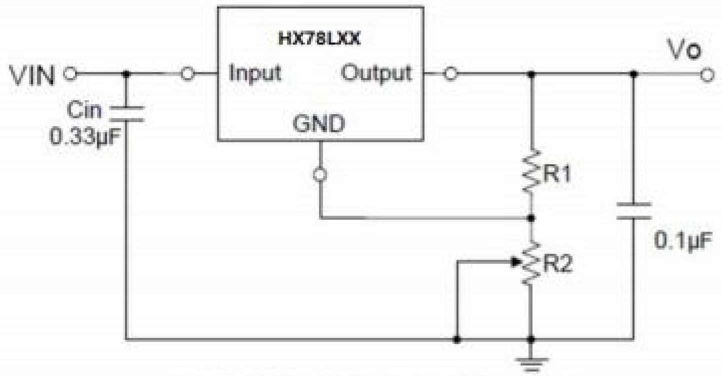


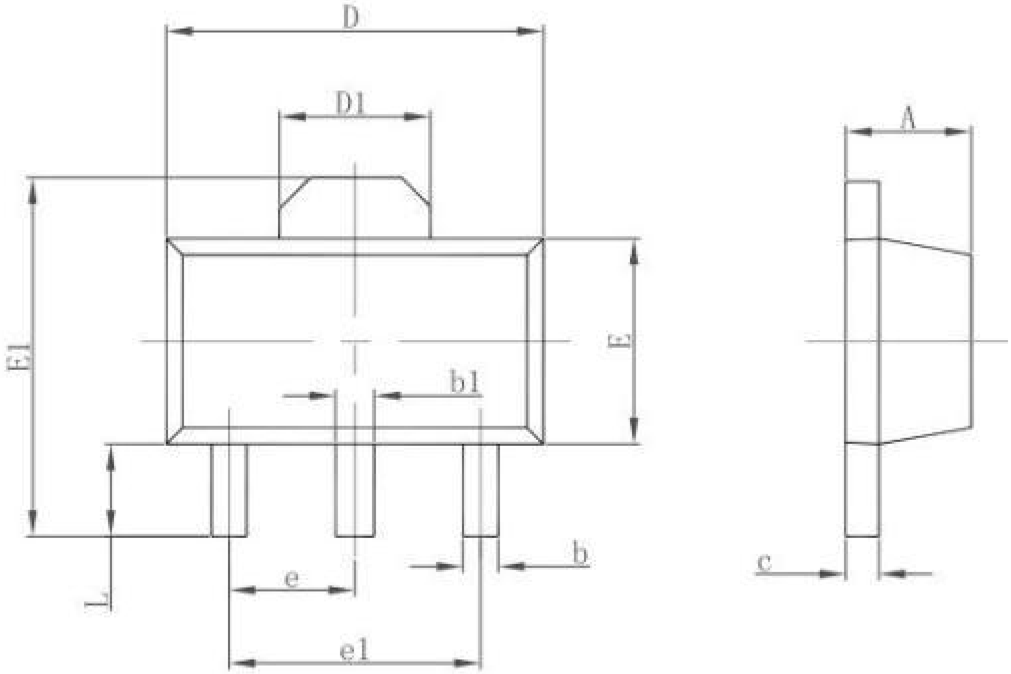
Fig.3 Adjustable Output Regulator

$$V_o=5V+(5V/R1+I_Q)*R2$$

$$5V/R1>3*I_Q$$

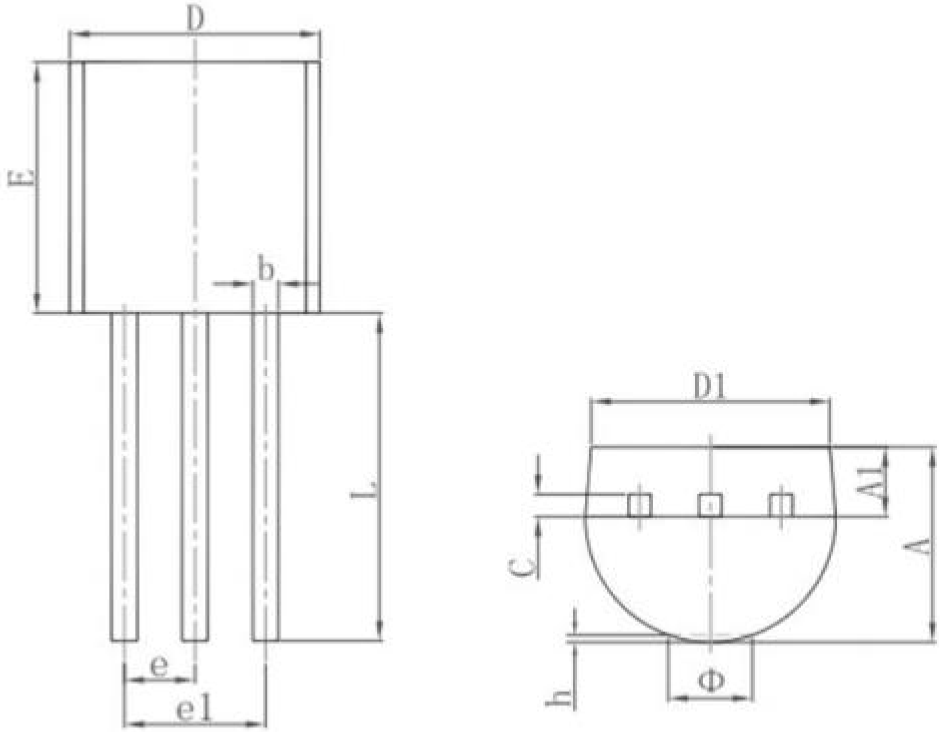
Package Information

3-pin SOT89 Outline Dimensions



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.400	0.580	0.016	0.023
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.550 REF.		0.061 REF.	
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
e	1.500 TYP.		0.060 TYP.	
e1	3.000 TYP.		0.118 TYP.	
L	0.900	1.200	0.035	0.047

3-pin TO92 Outline Dimensions



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	3.300	3.700	0.130	0.146
A1	1.100	1.400	0.043	0.055
b	0.380	0.550	0.015	0.022
c	0.360	0.510	0.014	0.020
D	4.300	4.700	0.169	0.185
D1	3.430		0.135	
E	4.300	4.700	0.169	0.185
e	1.270 TYP.		0.050 TYP.	
e1	2.440	2.640	0.096	0.104
L	14.100	14.500	0.555	0.571
Φ		1.600		0.063
h	0.000	0.380	0.000	0.015